# Recovery Certificates Under Bounded Provenance Omissions

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**Abstract**

An operator restoring an agent needs to know whether an incomplete dependency log could hide a path from compromised state to the restored service. We measure the smallest omission that invalidates a restoration certificate. Counting omitted edges is exact for an edge budget and can be arbitrarily optimistic when an entire log source can disappear. We give an exact source-based algorithm and an additive certificate that can sometimes settle the recovery decision without enumerating source combinations. On acyclic graphs, an exact algorithm stores only labels that can recur after the current vertex. The maximum number of such labels controls its state space. Separate source-subset enumeration checks both exact algorithms on synthetic graphs. An intervention can certify a logging source or rebuild an artifact. The two choices share a path-hitting condition, and optimizing their cost can be hard even when detecting a suspect path is simple.

## Summary and scope

**Question.** How many missing logging sources could conceal a path from compromised state to an artifact being restored?

**Finding.** The source-omission radius is the smallest number of distinct source labels along a suspect path. The exact search can forget labels that cannot recur, and an additive certificate can rule out a path under the declared source budget.

**Evidence and scope.** The result is proved and tested against source-subset enumeration. An incident team can use the certificate to decide whether it needs more evidence before restoring an artifact. The guarantee requires a complete candidate dependency graph.

## 1 The log that went quiet

Suppose an incident team rebuilds an agent's tool adapter and wants to restore a previously saved plan. The plan may still depend on compromised state through cached outputs. A missing log is especially important when it records several steps of one dependency chain. Counting those steps as several unrelated omissions gives a misleading picture of how much evidence must fail.

Let $G = (V, E_0 \cup E_*)$ be a finite directed candidate dependency graph after removing independently rebuilt artifacts from possible taint paths. The source $s$ represents suspect state and the target $t$ is an artifact whose recovery is proposed. Edges in $E_0$ are included in all modeled histories. Edges in $E_*$ are possible dependencies absent from the observed log. Each such edge has one source label $\ell(e)$. Activating a source permits all candidate edges with its label. We include all subsets of those permitted edges as possibilities. This models omission of an entire logging channel rather than loss of individual records.

The graph and its labels require validation outside this analysis. If a feasible dependency is absent even from $E_*$, or if a rebuild does not exclude suspect inputs, the model can miss a taint path. A certificate records the candidate graph, the assumed maximum number of affected sources, and the conditions that would invalidate either. Logs can justify an upper omission budget only through evidence external to the missing entries.

## 2 The minimum omission that opens a path

For an $s$-to-$t$ path $P$ write

$$m(P) = |P \cap E_*|, \qquad S(P) = \{\ell(e) : e \in P \cap E_*\}.$$

Take a minimum over the empty path family to be infinity.

**Theorem 1.** *If any set of at most $k$ optional edges can be absent from the log, a hidden taint path is possible exactly when $\min_P m(P) \le k$. If at most $k$ logging sources can be absent, the corresponding condition is $\min_P |S(P)| \le k$.*

*Proof.* A feasible hidden path uses only optional edges or source labels admitted by the omission set. Its cost is therefore at most the budget. Conversely, activate precisely the optional edges or labels of a path within the budget. The path is now present. □

We call the second minimum the source radius $r$. A known path gives $r = 0$. A proposed restoration is safe against the stipulated omission budget only if $r > k$. A chain of $L$ optional edges all recorded by one source has edge radius $L$ and source radius one. No constant factor can turn the edge radius into a lower certificate for the source radius.

The edge version is a shortest path with zero or unit costs. The source version is a minimum-color path with free observed edges. The difficulty of using few distinct colors is established in the graph literature [1]. In the directed setting, a direct reduction from set cover puts one layer boundary per universe element. Each arc crossing a boundary has a color corresponding to a set that contains that element. A path selects one such arc per element, and its distinct colors form a cover. Parallel arcs can be removed by inserting a private intermediate vertex and a free edge. Thus a polynomial exact algorithm is not expected in general.

## 3 A fast certificate and an exact fallback

Assume the candidate graph is acyclic, as in a dependency log ordered by creation. For each optional source $g$ let $M_g$ be the maximum number of its edges on any $s$-to-$t$ path. Compute $M_g$ by a longest-path pass with weight one on its edges and zero elsewhere. These passes use $O(g(|V|+|E|))$ operations for $g$ sources, apart from rational arithmetic. Ignore groups with $M_g = 0$. Give an optional edge labeled $g$ weight $1/M_g$ and every observed edge weight zero. On every path, each group contributes at most one, so the path's additive cost is at most $|S(P)|$. If $L$ is the minimum additive cost and $P_L$ attains it, then

$$\lceil L \rceil \le r \le |S(P_L)|.$$

The left inequality follows by minimizing a pointwise bound and rounding because $r$ is an integer. The right inequality comes from a concrete path. A candidate graph with no possible path can be recognized first by reachability.

If $\lceil L \rceil > k$, the bound certifies the restoration in the stated model. If $|S(P_L)| \leq k$, that witness disproves the guarantee. Otherwise use an exact algorithm. Keep, at each vertex, inclusion-minimal sets of source labels that reach it from $s$. Extend a set by taking its union with the label of the outgoing edge, then discard supersets. Removing a superset is safe because appending the same future labels preserves inclusion. Processing vertices in topological order yields the minimal sets at $t$, and their smallest cardinality is $r$. At most $2^g$ masks can occur when there are $g$ possible sources. For a decision at a fixed budget, sets exceeding that budget can be discarded early.

The certificate makes it possible to separate proof obligations. The cheap lower bound rules out all admitted hidden paths. The upper bound supplies a path and the exact sources whose absence would enable it. A recovery team can then look for independent evidence about those sources or rebuild an internal artifact on the witness path, and repeat the check on the amended graph. The resulting sequence is an auditable search for a sufficient repair. It does not by itself minimize the total repair cost.

## 4 Remembering only sources that can recur

The full antichain can contain labels that will never influence a later choice. At a vertex $v$ on some suspect path, let $B(v)$ be the union of optional labels on all $s$-to-$v$ paths and let $A(v)$ be the union on all $v$-to-$t$ paths. Call $L(v) = B(v) \cap A(v)$ the live labels. Two passes over a topological ordering compute these unions, ignoring vertices that cannot lie on an $s$-to-$t$ path.

**Theorem 2.** *For a prefix $P$ ending at $v$, store its full cost $|S(P)|$ and retain only the mask $S(P) \cap L(v)$. At every vertex, keeping the lowest cost for each mask preserves the exact source radius. If $w = \max_v |L(v)|$, the algorithm processes at most $O(2^w(|V| + |E|))$ states and transitions, with one mask operation and a table update per transition up to a constant factor.*

*Proof.* Suppose $P$ ends at $u$ with mask $M$. Extend it along $u \to v$. Let $Q$ be the singleton optional label of that edge, or the empty set for an observed edge. For any edge on an $s$-to-$t$ path, $Q \subseteq A(u)$. Consequently $Q \cap S(P) = Q \cap M$, and the cost increments by $|Q \setminus M|$. Also $A(v) \subseteq A(u)$, since a suffix from $v$ can follow the edge from $u$. Any previously used label that belongs to $L(v)$ was therefore already in $M$. The next mask is $(M \cup Q) \cap L(v)$. The increment and next mask depend only on the current mask, so a larger cost with that mask can be discarded. Start at $s$ with the empty mask and cost zero. Induction over vertices gives the minimum prefix cost for each reachable mask. At $t$, the suffix has no edges and the mask is empty, so its stored cost is the radius. Each table has at most $2^w$ entries. Iterating over its outgoing edges yields the stated bound, excluding bit-set and table-operation costs. □

This bound is useful when sources are numerous but each records a localized portion of a dependency history. Consider a chain of $k$ diamonds. Within each diamond, its two branches are each two optional edges of one label. Each diamond uses two labels that occur nowhere else. At a branch vertex the live width is one, and at a join it is zero. The full antichain at the target contains all $2^k$ choices, each with one label from every diamond. The live-label algorithm stores one cost per vertex, for $3k + 1$ states in total, and returns radius $k$.

## 5 Independent checks

An antichain solver and a second implementation enumerating activated sources agreed on all 256 seeded eight-vertex graphs. The additive lower and witness upper bounds agreed on 253. Counting

edges gave an optimistic answer about source robustness on three graphs. The separate seven-edge chain gave source radius one and edge radius seven.

To test harder instances, we used 256 ten-vertex graphs with five layers and seven possible sources. Every candidate path requires exactly four optional steps. Two solvers again agreed on all instances.

| Exact source radius | Instances | Lower and upper bounds agree |
|---:|---:|---:|
| 1 | 13 | 13 |
| 2 | 157 | 117 |
| 3 | 81 | 65 |
| 4 | 5 | 5 |

The inexpensive certificate settled the exact radius in 200 of 256 cases. The two bounds differed by one on 52 cases and by two on four. Using the edge radius four as the source radius would have overstated robustness on 251 cases. These counts describe the frozen graph generators and do not estimate real log failure frequencies. No timing speedup is claimed.

The live-label solver matched direct enumeration of activated sources on 512 additional acyclic graphs, including 76 with no possible path. It matched enumeration on diamond chains of one through five stages. On the 64-stage chain it returned radius 64 using 193 states, against the analytic count of $2^{64}$ terminal antichain entries. The latter antichain was counted from the construction without materializing it.

## 6 Choosing source audits and artifact rebuilds

Some source groups can be audited independently. A successful audit must establish that the group conceals no optional dependency in the current graph. An audit that instead discovers an edge requires updating the graph before the certificate can be used. Likewise, a rebuild removes the artifact from suspect paths only if it excludes all suspect inputs. Give each available audit and rebuild a cost.

Let $A$ be the selected audits and $R$ the selected rebuilds. A candidate path with at most $k$ distinct optional labels is an obligation. It is satisfied if its labels intersect $A$ or its internal artifacts intersect $R$. A path with neither available action cannot be certified away by these interventions.

**Theorem 3.** *The selected audits and rebuilds rule out all hidden paths under omission budget $k$ if and only if they satisfy every obligation. For audit-only plans with all labels auditable, budget two reduces to weighted vertex cover after selecting the forced singleton labels. Even at live width zero, deciding whether the cost of a sufficient audit plan fits a budget is NP-hard.*

*Proof.* Remove the optional edges ruled out by the audits and the vertices blocked by the rebuilds. If an original path with at most $k$ labels survives, activate exactly its optional labels. This is a feasible omission and gives a witness against restoration. Conversely, every hidden path in the residual graph has at most $k$ labels and survives precisely when neither intervention hit it.

With audits alone, a path with no optional labels makes the problem infeasible. At budget two, each singleton label is forced. Discard its satisfied obligations. Every remaining obligation has a pair of labels, so the remaining choice is a weighted vertex cover on the graph of these pairs. When this graph is bipartite, a minimum cut gives the optimum. Connect the source to left vertices and right vertices to the sink with their respective audit costs. Connect each adjacent left and right pair with capacity greater than the total available cost. An affordable cut chooses a left endpoint on the sink side or a right endpoint on the source side of each pair.

For the hardness claim, reduce ordinary vertex cover [2]. Make a label for each vertex. For each graph edge use a separate two-arc route from $s$ to $t$ whose two labels are its endpoints. All such paths have two labels. No branch repeats its first label, so every live set is empty. At budget two the audit plan must cover exactly the original edges. □

An exact optimizer can find one residual witness, branch on every eligible audit or rebuild that hits it, and retain the cheapest valid completion. The live-label solver with predecessor pointers can supply the witness. A witness with no eligible actions closes that branch as infeasible. This separates the cost of finding one counterexample from the cost of choosing the intervention set. For an audit-only plan with at most $b$ interventions, each branch has at most $k$ children and depth at most $b$.

As a small example, two suspect routes depend on source one and a third depends on source two. Their distinct intermediate artifacts cost $2, 2, 1$ to rebuild. The audits cost 1 and 4. Auditing source one and rebuilding the third artifact costs 2. Using either intervention type alone costs 5. The witness solver matched full intervention and omission enumeration on 192 seeded graphs, and matched weighted vertex cover on 48 additional zero-width fixtures. These are checks of the declared graph model. A recovery team can use the mixed plan when an independent audit is cheaper than rebuilding several artifacts. The certificate must record what the audit established about coverage.

## 7 What a recovery drill should measure

Record source-level outage groups and keep the label meaning stable across tools. Inject a known complete log into a recovery drill, withhold selected records or whole sources, and require the team to propose a restoration before seeing the hidden graph. Then test whether the actual hidden graph fits the candidate family, whether the claimed budget held, and whether the selected artifacts still receive taint. Measure unnecessary rebuilds, time to safe service, and residual taint separately. If common infrastructure can hide several sources together, represent that shared failure in the budget model before using the certificate.

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## References


[1] Hajo Broersma, Xueliang Li, Gerhard J. Woeginger, and Shenggui Zhang. *Paths and Cycles in Colored Graphs*. Australasian Journal of Combinatorics 31, 299–311, 2005. `https://ajc.maths.uq.edu.au/pdf/31/ajc_v31_p299.pdf`.

[2] Richard M. Karp. *Reducibility Among Combinatorial Problems*. Complexity of Computer Computations, 1972. `https://www.cs.umd.edu/~gasarch/BLOGPAPERS/Karp.pdf`.